10. Which of the following statements is wrong regarding conduction of nerve impulse?

    a. In a resting neuron, the axonal membrane is more permeable to K^+ ions and nearly impermeable to Na^+ ions.

    b. Fluid outside the axon has a high concentration of Na^+ ions and low concentration of K^+, in a resting neuron.

    c. Ionic gradient s are maintained by Na^+ K^+ pumps across the resting membrane, which transport 3Na ions outwards for $2K^+$ into the cell.

    d. A neuron is polarized only when the outer surface of the axonal membrane possess a negative a charge and its inner surface is positively charged.

11. All of the following are associated with the myeline sheath except

    a. Faster conduction of nerve impulses

    b. Nodes of Ranvier forming gaps along the axon

    c. Increased energy output for nerve impulse conduction

    d. Saltatory conduction of action potential

12. Several statements are given here in reference to cone cells which of the following option indicates all correct statements for cone cells ?

    Statements

    (i) Cone cells are less sensitive in bright light than Rod cells

    (ii) They are responsible for colour vision

    (iii) Erythropsin is a photo pigment which is sensitive to red colour light

    (iv) They are present in fovea of retina

    a. (iii), (ii) and (i)

    b. (ii), (iii) and (iv)

    c. (i), (iii) and (iv)

    d. (i), (ii) and (iv)

13. Which of the following statement concerning the somatic division of the peripheral neural system is incorrect?

    a. Its pathways innervate skeletal muscles

    b. Its pathways are usually voluntary

    c. Some of its pathways are referred to as reflex arcs

    d. Its pathways always involve four neurons

14. When the potential across the axon membrane is more negative than the normal resting potential, the neuron is said to be in a state of

    a. Depolarization

    b. Hyperpolarization

    c. Repolarization

    d. Hypopolarization

15. Why is the blind spot called so?

16. Sam's optometrist tells him that his intraocular pressure is high. What is this condition called and which fluid does it involve?

17. The action potential occurs in response to a threshold stimulus; but not at sub threshold stimuli. What is the name of the principle involved?

18. Pleasant smell of food urged Ravi to rush into the kitchen. Name the parts of the brain involved in the identification of food and emotional responses to odour.

19. Cornea transplant in humans is almost never rejected. State the reason.

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